

Business Analytics Capstone

MIS 382

General Motors

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Assignment # 5

Reflective Journal:

1. Key learnings and Insights:

- Understanding EV adoption trends and key factors influencing infrastructure planning, such as traffic data, vehicle registration, and existing charging station networks.
- Importance of financial modeling, including ROI calculations and forecasting charger utilization revenue.
- Realizing the significance of accurate data merging, as GM's dataset lacked updated demographic and sales information.
- Exposure to open-source datasets and their usability for validating business assumptions.
- Experience in handling real-world project constraints, such as dealing with incomplete or delayed datasets.

2. High Points for the team:

- Successfully set up a structured 13-week project plan, ensuring clear milestones and deliverables.
- Developed an ROI model with a positive NPV of \$0.3M and a payback period of 3.8 years, demonstrating the financial viability of the project.
- Established a GitHub repository to organize documentation and track version history.
- Completed accurate ZIP code mapping for traffic data, allowing for better geographic insights.

3. Low Points for the team:

- The most updated GM dataset was received only shortly before client meetings, limiting the time available for analysis.
- Some critical GM-provided data fields were outdated or missing (e.g., charging station details, demographic updates).
- Difficulty in obtaining EV-specific financial metrics from sources like Capital IQ, requiring assumptions based on industry benchmarks.
- Challenges in matching disparate datasets (e.g., GM sales data with external traffic and EV registration data).

4. Project Plan adjustment:

- Adapted data strategy by leveraging external sources (e.g., Tennessee traffic and EV adoption data) to compensate for missing GM data.
- Implemented a phased approach to the dataset integration, prioritizing high-impact data points first.

5. Main Challenges:

- Data delays and inconsistencies: The late arrival of updated datasets required the team to make interim assumptions.
- Granularity issues: Traffic data lacked ZIP codes, requiring additional steps to map coordinates to ZIP codes.
- Financial modeling complexity: Lack of EV-specific COGS breakdown meant relying on industry-wide estimates.
- Infrastructure cost variations: Rent costs for charging stations varied significantly across urban, suburban, and rural areas, requiring classification efforts.
- Data integration: Merging GM data with third-party datasets was challenging due to differences in granularity and missing attributes.